

TAO LIN

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EDUCATION

Zhejiang University

B.Eng. in Computer Science and Technology

October 2012 - Present

Hangzhou, China

Overall GPA: 3.96/4.00 (92.45/100)
The third year GPA: 4.00/4.00 (94.43/100)
Rank: 1st in Qizhen Honors Class in Chu Kochen Honors College
Top 2% among 196 students

The Hong Kong University of Science and Technology

Exchange Student for Capstone Project

September 2015 - February 2016

Hong Kong

EXPERIENCE

Visual Analytics Group, State Key Lab of CAD&CG

Research Assistant

June 2013 - August 2015

Zhejiang University, Hangzhou, China

Mentor: Professor Wei Chen [[Homepage](#)]

- **Administrated** a computer cluster with 20 nodes. **In charge of** processing large-scale data on the cluster with Hadoop and Spark.
- Designed and implemented a visual analytics system supporting situation-aware understanding and visual reasoning of human mobility. (See Project Mobility Viewer)
- Designed and **predominantly** implemented a new visual analytics framework and system that enables interactive analysis and exploration of the dynamic changes of interpersonal ties. (See Project TieVis)

Multimedia Technology Research Center

Visiting Intern

September 2015 - Present

Hong Kong University of Science and Technology, Hong Kong

Mentor: Professor Gary Chan [[Homepage](#)]

- Work for Streamphony, a next-generation streaming cloud for large-scale high bit-rate applications over the global Internet. Developing the research results into a practical system for commercial trials
- **Independently** implement, test and improve the Forward Error Correction (FEC) Module.

PUBLICATION

Mobility Viewer: A Eulerian Approach for Studying Urban Crowd Flow

Yuxin Ma, **Tao Lin**, Zhendong Cao, Chen Li, Wei Chen

IEEE Transactions on Intelligent Transportation Systems. Accept, 2015. [[PDF](#)] [[Video](#)]

- A suite of interactive visualization techniques for exploring crowd flow volumes, directional dynamics and spatiotemporal patterns.
- A visual analysis approach for inspecting human social relations and their physical movements.

TieVis: Visual Analytics of Evolution of Interpersonal Ties

Tao Lin, Fangzhou Guo, Yingcai Wu, Wei Chen, Biao Zhu, Sidong Wang, Huamin Qu

The 9th IEEE Pacific Visualization Symposium (PacificVis 2016). Submitted. [[PDF](#)] [[Video](#)] [[Code](#)]

- A new study of the evolution of interpersonal ties for a dynamic network and their co-evolution with the network structure and information diffusion.
- An edge based analysis framework which helps users identify edges with similar trends, compare edges with different trends, and find the hidden patterns.

HONORS AND AWARDS

Google Excellence Scholarship <i>58 Bachelor and Master students in mainland China</i>	2015 <i>Google Inc.</i>
National Scholarship <i>Top 1% students</i>	2014 - 2015 <i>Ministry of Education, China</i>
First Prize of Excellent Undergraduate Scholarship <i>Top 3% students</i>	2013 - 2014, 2014 - 2015 <i>Zhejiang University</i>
Honorable Mention, Interdisciplinary Contest In Modeling	2015
Second Prize, Zhejiang University Programming Contest	2014
Third Prize, ACM-ICPC China Zhejiang Provincial Programming Contest	2014

SELECTED PROJECTS

Mobility Viewer May 2014 - June 2015
Core Member [\[PDF\]](#) [\[Video\]](#) *Visual Analytics Group, Zhejiang U.*

- **Independently** processed 14 billion trajectory records (about 1.8 terabytes) and 4.5 million call detailed records on a computer cluster with 20 nodes by Spark. The processing includes (1)cleaning missing, invalid and duplicated records, (2)detecting and eliminating ping-pong effects by a newly designed algorithm, and (3)slicing and aggregating for later use.
- **Independently** designed and implemented the Flow Volume Analysis Subsystem, containing a sunburst view and a 3-D terrain view, by JavaScript (D3.js, three.js) on OpenSteetMap.

TieVis May 2015 - October 2015
Project Leader [\[PDF\]](#) [\[Video\]](#) [\[Code\]](#) *Visual Analytics Group, Zhejiang U.*

- Designed and **predominantly** implemented TieVis, an interactive visualization system that integrates five views, including a scatterplot, a pixelbar chart, a layered graph, a node-link diagram and a information panel, which allows for multi-perspective exploration and analysis.
- Preprocessed the data by Python, including extracting, aggregating and calculating PCA and MDS.

Forward Error Correction Module in Streamphony September 2015 - Present
Visiting Intern *Multimedia Technology Research Center, HKUST*

- **Independently** implement Forward Error Correction (FEC) Module by C++11, using STL and Boost (asynchronous, memory pool and smart pointer) heavily. This module could encode and decode (recover) the video stream by Reed-Solomon error correction.
- Cooperate with tens of software engineers in Streamphony. Do the code review and project management by Phabricator, and write unit tests for all codes by Google Test and Google Mock.

Visual Analysis of Library Data April 2014 - May 2015
Team Leader [\[Video\]](#) [\[Code\]](#) *Student Research Training Program*

- It is a visual analysis system helps readers and administrators in the library make decisions by interacting with real records.
- **Predominantly** designed and implement the system, a single-page application (SPA). The front-end uses JavaScript libraries including D3, Bootstrap, ECharts and jQuery. The back-end uses Django by Python, and connect the real business Oracle database.

Precomputed Shadow Fields for Dynamic Scenes March 2015 - July 2015
Independent [\[Code\]](#) *Course Project*

- Final project for the course Advances in Computer Graphics, Score: 94, Instructor: Kun Zhou.
- A realization of Precomputed Shadow Fields for Dynamic Scenes, a paper in SIGGRAPH 2005.
- Implemented by DirectX and HLSL, with knowledge about rendering, including precomputed radius transfer, dynamic soft shadows and spheric harmonics function.

A 3D Auto-Stereoscopic Front Projection System July 2014 - May 2015
Programmer [\[Video\]](#) *Challenge Cup Competition*

- The system adopts the light field reconstruction principle to achieve 3D views with unaided eyes. It employs the unique front-projection configuration, which makes the system cover less space.
- Developed with 5 students major in optical engineering, computer science, and architecture.
- Derived the core mathematical formulas and implemented the prototype with OpenGL.

TECHNICAL STRENGTHS

Computer Languages	C/C++, JavaScript, Python, Java, MATLAB, Verilog
Libraries & Frameworks	Visualization (D3, ECharts) C++ Unit Test (Google Test, Google Mock)
Tools	Computer Graphics & Vision (OpenGL, DirectX, OpenCV) Data Science (Hadoop, Spark, Hive, MySQL) Software Engineering (Unix Command, Git, Phabricator)